

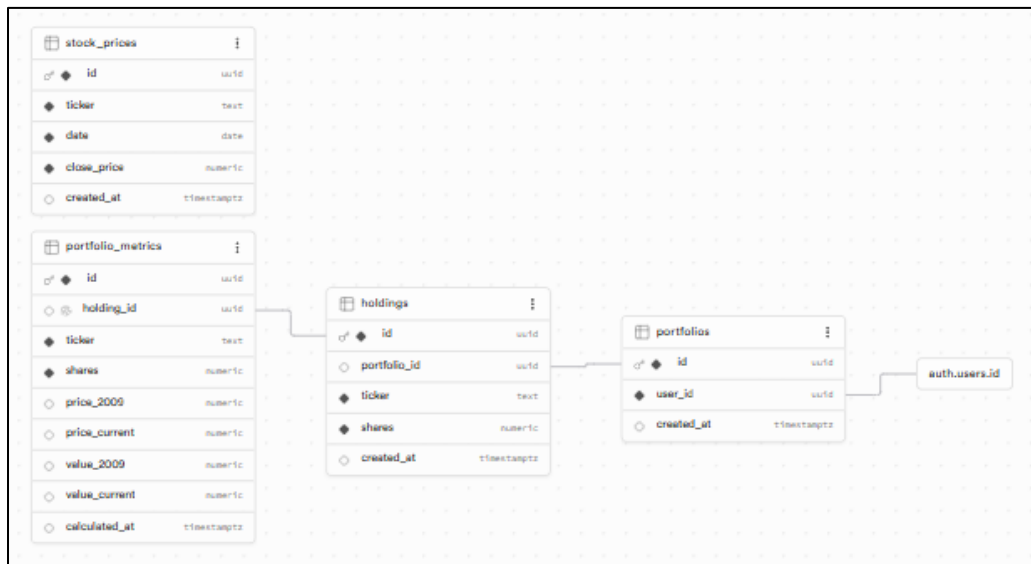
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MSBA 692: Pipelines to Insights

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Schema Documentation: The “Should I Sell?” Stock Dash Tool

Stock Analytics Data Warehouse



This database in Supabase stores user portfolio data and historical stock price information, enabling the calculation of performance metrics for individual holdings.

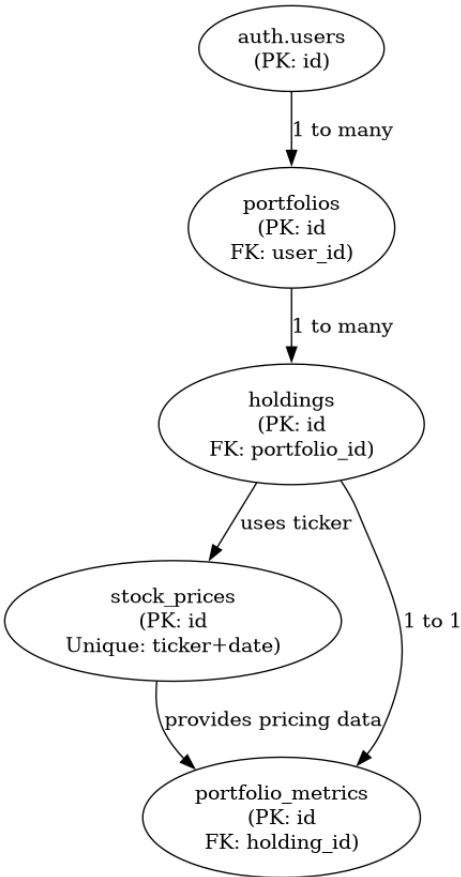
The system is designed to support a data pipeline where:

- Users input ticker and share count of a stock holding
- External API provides historical and current price data
- Computed metrics are stored for fast dashboard access

The schema is normalized to approximately Third Normal Form (3NF) by:

- Separating user, portfolio, and market data into distinct tables
- Avoiding redundant storage of stock price information
- Deriving metrics from normalized tables and storing them for performance

ER Diagram



Database Overview

The database contains four core tables:

- Portfolio
- Holdings
- Stock_prices
- Portfolio_metrics

Entity Relationship Summary

Table	Purpose
portfolios	Stores user portfolio containers
holdings	Stores individual stock positions (ticker + shares)
stock_prices	Stores historical and current stock price data
portfolio_metrics	Stores precomputed portfolio performance metrics

Table Documentation

1. Portfolios

a. Purpose

- i. Stores portfolio recrds associated with users. Each user can have one or more portfolios

b. Primary Key

- i. Id

c. Foreign Keys

- i. User_id --> auth.users.id

d. Relationships

- i. One user can have many portfolios
- ii. One portfolio can have many holdings

e. Table Structure

Column Name	Data Type	Key	Description
id	UUID	Primary Key	Unique portfolio identifier
user_id	UUID	Foreign Key	Reference to authenticated user
created_at	TIMESTAMP		Portfolio creation timestamp

2. Holdings

a. Purpose

- i. Stores individual stock holdings within a portfolio; each record represents a stock ticker and the number of shares owned.

b. Primary Key

- i. Id

c. Foreign Keys

- i. portfolio_id --> portfolios.id

d. Relationships

- i. Many holdings belong to one portfolio
- ii. Each holding maps to many historical price records via stock_prices
- iii. Each holding has one corresponding metrics record in portfolio_metrics

e. Table Structure

Column Name	Data Type	Key	Description
id	UUID	Primary Key	Unique holding identifier
portfolio_id	UUID	Foreign Key	Parent portfolio
ticker	TEXT		Stock symbol (e.g., AAPL, MSFT)
shares	NUMERIC		Number of shares held
created_at	TIMESTAMP		Record creation timestamp

3. Stock_prices

a. Purpose

- i. Stores historical and current stock price data retrieved from external API. This table supports historical comparisons, current valuation, and future performance analytics potentials.

b. Primary Key

- i. Id

c. Constraints

- i. Unique combination of (ticker, date)

d. Relationships

- i. Many price records belong to one ticker
- ii. Used to compute portfolio metrics

e. Table Structure

Column Name	Data Type	Key	Description
id	UUID	Primary Key	Unique price record identifier
ticker	TEXT		Stock symbol
date	DATE	Unique (with ticker)	Price date
close_price	NUMERIC		Closing price for the day
created_at	TIMESTAMP		Data ingestion timestamp

1. Portfolio_Metrics

a. Purpose

- i. Stores precomputed performance metrics for each holding. This table is designed to improve query performance for dashboards, avoid repeated joins across large datasets, and provide fast access to historical and current valuations.

b. Metrics Included

- i. Value at bottom of the Great Recession (June 2009), current value, absolute return, percentage return

c. Primary Key

- i. Id

d. Constraints

- i. One metrics record per holding (UNIQUE holding_id)

e. Relationships

- i. One holding has one metrics record
- ii. Metrics are recalculated when holdings or price data changes

f. Table Structure

Column Name	Data Type	Key	Description
id	UUID	Primary Key	Unique metrics record
holding_id	UUID	Foreign Key	Associated holding
ticker	TEXT		Stock symbol
shares	NUMERIC		Number of shares
price_2009	NUMERIC		Closing price (June 2009 benchmark)
price_current	NUMERIC		Latest available price
value_2009	NUMERIC		Value at 2009 bottom (shares × price_2009)
value_current	NUMERIC		Current value (shares × price_current)
return_absolute	NUMERIC		Profit/loss since 2009
return_percent	NUMERIC		Percentage return

calculated_at	TIMESTAMP		Last calculation timestamp
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Cardinality Relationships

Parent Table	Child Table	Relationship Type
auth.users	portfolios	One-to-Many
portfolios	holdings	One-to-Many
holdings	portfolio_metrics	One-to-One
stock_prices	portfolio_metrics	Derived Relationship

Normalization Notes (3NF)

This schema is normalized to approximately Third Normal Form:

- Portfolio data is separated from holdings
- Stock price data is centralized in stock_prices to avoid duplication
- Metrics are derived from normalized data rather than stored redundantly in base tables
- Each non-key attribute depends only on its table's primary key
- Derived metrics are stored in a separate table for performance optimization

Data Sources

Stock price data is retrieved from external API: Alpha Vantage API. Other options could be polygon.io, or Yahoo Finance if alternatives are needed.

Example Relationship Flow

1. A user creates a portfolio
2. The user adds a holding (i.e. ticker = AAPL, shares = 10)
3. The system retrieves AAPL price on June 1, 2009, and current price.
4. Data is stored in stock_prices, and portfolio_metrics

Architecture Overview

Users → Portfolios → Holdings → Stock Prices → Portfolio Metrics → Dashboard